Quantitative Research

Part 1: Descriptive Statistics

Prof. Dr. Christoph Weisser

HSBI

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About the Lecturer



Prof. Dr. rer. pol. Christoph Weisser

MLitt (St Andrews), MSc (Oxford)

Professor of Mathematics, specializing in Business Data Science, at HSBI

- Former Technical Lead for Artificial Intelligence and supervisor of industrial doctoral projects at BASF
- Research: Data Science, Machine Learning and Artificial Intelligence
- Focus: Application of AI in business and industry

Contact: christoph.weisser@hsbi.de

The lecture at a glance

	Part	Content
▶	**1 – Descriptive Statistics**	**Basic concepts, levels of measurement, frequencies, measures of central tendency and dispersion**
	2 – Graphical Representations	Empirical CDF, histogram, boxplot
	3 – Statistical Inference	Confidence intervals, hypothesis tests, p-values, one-sided tests, types of error
	4 – Regression	Correlation, simple and multiple linear regression, inference for regression, diagnostics
	5 – Classification	Logistic regression, classification rules, confusion matrix, ROC/AUC, Bayes classifiers

All analyses are carried out in **R** (with RStudio); every deck comes with an R script that reproduces its numbers.

Software and References: R and RStudio

R. M. Venables, D. M. Smith and the R Development Core Team. *An Introduction to R*.



Contents – Part 1

1. Fundamentals
2. Descriptive Statistics
3. Summary

Learning objectives – Part 1

After this part you will be able to ...

- name the **basic concepts** (statistical unit, variable, value, observation) and the **levels of measurement**,
- compute **absolute, relative and cumulative frequencies**,
- compute and interpret **measures of central tendency** (means, median, mode, quantiles),
- compute and compare **measures of dispersion** (s^2, s, v, v^*),
- choose the **right measure** for a given comparison,
- carry out every computation in R.

Key takeaway

One running story: the **shoe sizes of a soccer team** – we describe this small data set completely, first in numbers (Part 1), then in pictures (Part 2).

List of Symbols (1)

Symbol	Meaning	Symbol	Meaning
ω	Statistical unit	h_j	Absolute frequency
n	Number of statistical units	f_j	Relative frequency
X	Variable	H_j	Cumulative absolute frequency
x	Value (category)	$F(x)$	Empirical distribution function
x_i	Observation	x_j	Value (class j)
$x_{(i)}$	i-th value of the ordered sample	$\tilde{x}$	Median
$\bar{x}$	Arithmetic mean	$\bar{x}_{\text{geo}}$	Geometric mean

List of Symbols (2)

Symbol	Meaning	Symbol	Meaning
s^2	Variance (sample)	v	Coefficient of variation
s	Standard deviation	v^*	Normalized coefficient of variation
R	Range	x_p	p-quantile
IQR	Interquartile range	$x_{0.25}$, $x_{0.75}$	25 %- and 75 %-quantile
$\lceil \cdot \rceil$	Rounding up to an integer	Q_1, Q_3	1st and 3rd quartile

Symbols for regression (s_{xy}, r, β_0, β_1, R^2, ...) are introduced in Part 4.

Overview

1 Fundamentals

2 Descriptive Statistics

3 Summary

Basic Concepts

Statistical unit (ω): Individual object being observed. Must be defined in terms of subject matter, time, and location. e.g. daily maximum temperature, on which day, at which location

n: The number of statistical units

Sample: The collected set of statistical units

Population: The overarching set from which the sample was drawn

Basic Concepts: Variable and Observed Values

Variable (X): A characteristic of the statistical unit. There may be several characteristics; which one is examined must be specified.

Values (x): Possible values that the variable can take.

Observation (x_i): The value that was actually observed for the variable.

Basic Concepts – Summary

Term	Symbol	Explanation	Example
Statistical unit	ω	Unit of analysis	Persons, institutions, objects
Variable	X	Characteristic of interest of a statistical unit	Age, market value, weight, height, costs, rating, product count, color
Value	x	Values, states, categories that a variable can take	Euros, kg, cm, well-being, grades, yes or no, black, white, blue, ...
Observation	x_i	The actually observed value of a variable	31 years, 1300 euros, 23 cm, 73 kg, black

Data Collection and Survey Types

Data Collection

- **Primary data:** Data collected specifically for this study
- **Secondary data:** Data taken from existing statistics

Survey Types

- Observation
- Interview / Questionnaire
- Experiment

Data Protection

- Regulated by law, e.g. the GDPR
- Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data, on the free movement of such data, and repealing Directive 95/46/EC
- As well as further national legislation

Important

Compliance with data protection is a prerequisite for data analysis!

Levels of Measurement 1

Term	Explanation	Example
Nominal scale	Status/category only, no ordering	Color, gender, origin, occupation
Ordinal scale	Ranking possible, differences not meaningful	Attitudes, well-being, grades
Metric scale	Ranking possible, differences meaningful	Monetary units, temperatures

Levels of Measurement 2

- Categorical data includes nominal and ordinal data.

Important

A metric level of measurement is a prerequisite for most of the statistical analyses covered in this course!

Overview

1 Fundamentals

2 Descriptive Statistics
- Frequencies
- Measures of Central Tendency
- Measures of Dispersion

3 Summary

Frequencies

Frequencies – Absolute, Relative, Cumulative

Frequencies

Absolute frequency: How often does a value occur:

$$h_j = \#\{i \mid x_i = x_j\}$$

Relative frequency: Proportion of the total count:

$$f_j = \frac{h_j}{n}$$

Cumulative relative frequency: Accumulated relative frequencies:

$$F(x) = \sum_{x_j \leq x} f_j$$

The Data – an Example

The statistical units: the players of a soccer team



The Data – an Example 2

The variable: The shoe sizes of a soccer team with the following observations:

40, 39, 38, 41, 46, 39, 39, 38, 37, 40, 39

Frequency Table

Shoe size	37	38	39	40	41	46	Total
absolute frequency	1	2	4	2	1	1	11
relative frequency (decimal)	0.0909	0.1818	0.3636	0.1818	0.0909	0.0909	1
relative frequency in %	9.1 %	18.2 %	36.4 %	18.2 %	9.1 %	9.1 %	100 %
relative frequency (fraction)	$\frac{1}{11}$	$\frac{2}{11}$	$\frac{4}{11}$	$\frac{2}{11}$	$\frac{1}{11}$	$\frac{1}{11}$	1
cumulative rel. freq. (decimal)	0.0909	0.2727	0.6364	0.8182	0.9091	1	
cumulative rel. freq. (fraction)	$\frac{1}{11}$	$\frac{3}{11}$	$\frac{7}{11}$	$\frac{9}{11}$	$\frac{10}{11}$	1	

Why Statistics Software?

Advantages of Modern Software

- Complex calculations in seconds
- Reproducible analyses through scripts
- Professional visualizations
- Ability to handle large datasets

In this course: We use **R** and **RStudio**



www.r-project.org

What is R?

- **R** is a programming language for statistical analysis and data analysis.
- Typical application areas:
  - Data Science
  - Statistics
  - Machine Learning
  - Scientific research
- Strengths of R:
  - Excellent data visualization (e.g. `ggplot2`)
  - Large selection of packages
  - Comprehensive statistical methods
- **Important:** R is the *computation engine* (performs calculations).

What is RStudio?

- **RStudio** is an integrated development environment (IDE) for R.
- It provides a user-friendly interface:
  - Script editor
  - Console for executing code
  - Window for plots and visualizations
  - Package management
- **Analogy:**
  - R = Engine of a car
  - RStudio = Dashboard and steering wheel
- **Important:** RStudio only works if R is installed.

Installation (Mac & Windows)

- **Step 1: Install R**
  - Website: `https://cran.r-project.org`
  - Download:
    - Windows: "Download R for Windows"
    - Mac: "Download R for macOS"
  - Follow the installation instructions
- **Step 2: Install RStudio**
  - Website: `https://posit.co/download/rstudio-desktop/`
  - Download **RStudio Desktop (free)**
  - Install like a regular program
- **Step 3: Launch RStudio**
  - RStudio automatically detects the R installation
  - After that, you can start working with R directly

Frequencies with R

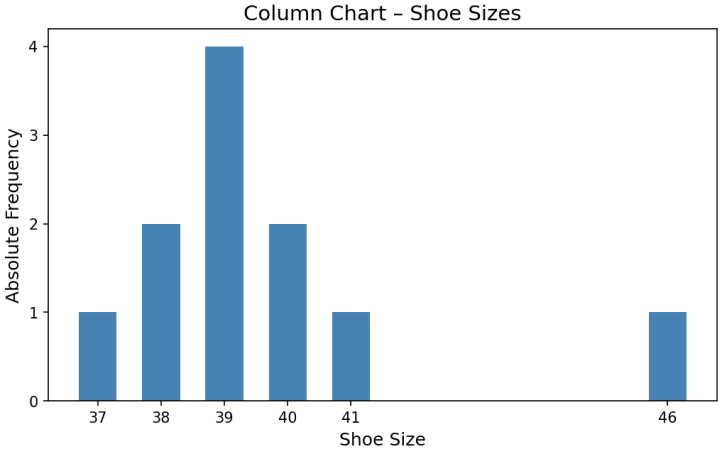
```r
# Soccer team - shoe sizes of the 11 players
data ← c(40, 39, 38, 41, 46, 39, 39, 38, 37, 40, 39)

# Frequency table
table(data)

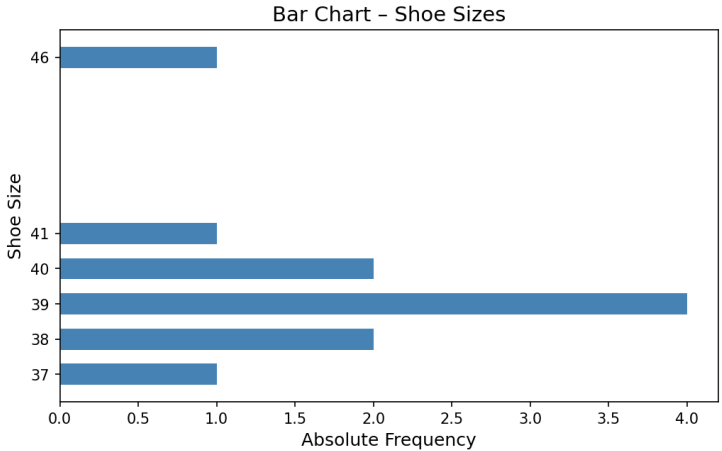
# Relative frequencies
prop.table(table(data))

# Cumulative frequencies
cumsum(prop.table(table(data)))
```

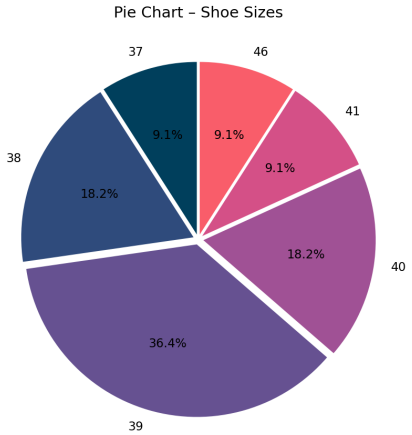
Column Chart



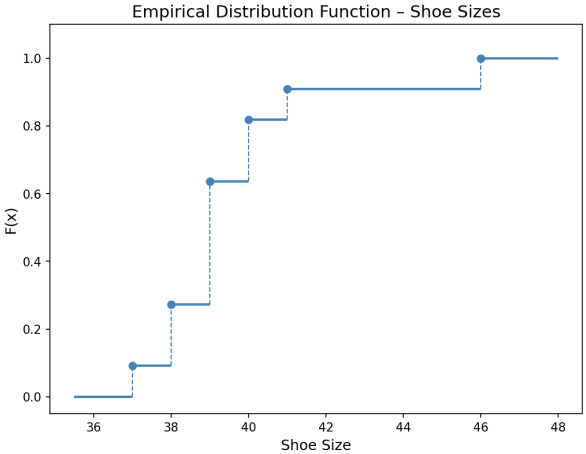
Bar Chart


Bar Chart – Shoe Sizes
Shoe Size
46
41
40
39
38
37
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Absolute Frequency

Pie Chart



Empirical Distribution Function



Measures of Central Tendency

Measures of Central Tendency – Means, Median and Quantiles

Measures of Central Tendency

- Where is the center?
- Where is the center of the distribution?
- At which value is one quarter of the values reached?
- At which value are three quarters of the values reached?

Means

Arithmetic mean: Sum of all values divided by n.

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

Geometric mean: Product of all values, n-th root.

$$\bar{x}_{\text{geo}} = \left(\prod_{i=1}^{n} x_i \right)^{1/n}$$

Recommended for growth rates.

- **Advantage:** easy to compute
- **Disadvantage:** sensitive to extreme values (outliers)
- **Trimmed mean:** Extreme values removed, remainder $\rightarrow$ arithmetic mean

Median

Definition

Below which observation are at least 50 % of the values?

Computation: $\lceil n/2 \rceil$ = position of the median in the sorted values

Note: alternative definition

In some textbooks, the median for even n is defined as the average of the two middle values (the "middle value"). In this course, we use the $\lceil n/2 \rceil$ definition.

Advantages of the $\lceil n/2 \rceil$ definition:

- The median is always an actual observation
- Consistent with the quantile definition: Median = 50 %-quantile = $x_{\lceil 0.5 \cdot n \rceil}$

Advantages of the median in general:

- Robust to outliers
- Also suitable for ordinal data

Mode

Definition

The most frequent value in the distribution

- **Advantage:** suitable for both ordinal and nominal data
- **Disadvantage:** often not unique
- **Disadvantage:** may not represent the "center" of the distribution

Measures of Central Tendency – Example

Player	1	2	3	4	5	6	7	8	9	10	11
Shoe size	40	39	38	41	46	39	39	38	37	40	39

Sum	436
Arithmetic mean	39.6364
Geometric mean	39.5755
Trimmed mean	39.2222
Median	39
Mode	39

Measures of Central Tendency with R 1

```r
# Sum
sum(data)

# Arithmetic mean
mean(data)

# Geometric mean
# Step 1: Calculate the product of all values
product ← prod(data)
# Step 2: Determine the number of observations
n ← length(data)
# Step 3: Take the n-th root
gmean ← product^(1/n)
```

Measures of Central Tendency with R 2

```r
# Sum
sum(data)

# Trimmed mean: 10% of extreme values are removed
mean(data, trim = 0.10)

# Median
median(data)  # Note: R's definition may differ!
```

Quantiles

First, the values must be sorted!

1st quartile: Below which value are at least 25 % of the values?

2nd quartile: = Median (Below which value are at least 50 % of the values?)

3rd quartile: Below which value are at least 75 % of the values?

Arbitrary p%-quantile: Below which value are at least p % of the values?

Quantiles – Computation

For the p-quantile: round up $n \cdot p$, position in sorted values

Example 80%-quantile: $n \cdot 0.8 = 11 \cdot 0.8 = 8.8$, round up to position 9.

Quantiles – Computation Example

Quantile	p	$n \cdot p$	Position	Value
1st quartile	0.25	2.75	3	38
2nd quartile	0.50	5.50	6	39
3rd quartile	0.75	8.25	9	40
17%-quantile	0.17	1.87	2	38
80%-quantile	0.80	8.80	9	40
5%-quantile	0.05	0.55	1	37

Quantiles – Worked Example

Sorted data: 37, 38, 38, 39, 39, 39, 39, 40, 40, 41, 46

$$\text{Median} = x_{\lceil 0.5 \cdot 11 \rceil} = x_6 = 39$$
$$Q_1 = x_{\lceil 0.25 \cdot 11 \rceil} = x_3 = 38$$
$$Q_3 = x_{\lceil 0.75 \cdot 11 \rceil} = x_9 = 40$$

Quantiles with R

```r
# All standard quantiles at once (type = 1: discrete)
quantile(data, type = 1)

# 1st quartile (25%-quantile)
quantile(data, 0.25, type = 1)

# 80%-quantile
quantile(data, 0.80, type = 1)
```

Measures of Dispersion

Measures of Dispersion – Variance, Standard Deviation, Coefficient of Variation, Normalized Coefficient of Variation

Measures of Dispersion – Why?

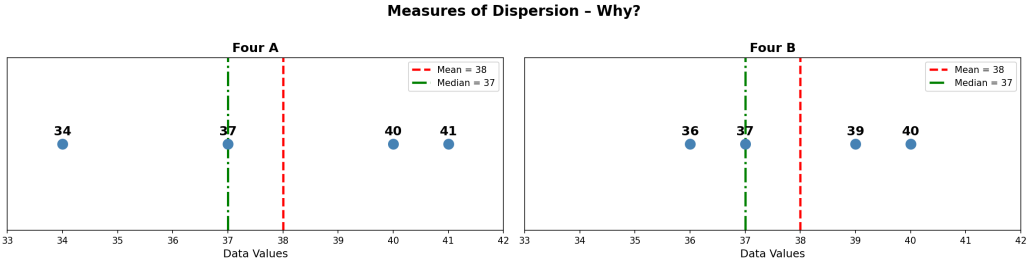
Example: Shoe sizes of two women's rowing fours (without coxswain)

	Four A	**Four B**
Data	34, 37, 40, 41	36, 37, 39, 40
Mean	38	38
Median	37	37

Question

The mean and median are identical. Does that mean the data sets are equivalent?

Measures of Dispersion – Visualization



- Mean and median are identical – but the **dispersion** differs considerably
- Four A: Data farther from the mean $\rightarrow$ greater dispersion
- Four B: Data closely concentrated around the mean $\rightarrow$ less dispersion

Sample Variance

Definition

Mean squared deviation from the mean:

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2 \qquad 0 \leq s^2 \leq \infty$$

- **Why square?** – So that positive and negative deviations from the mean do not cancel out!
- **Why divide by $n-1$ only?** – Because using the mean leaves only $n-1$ degrees of freedom.
  - If the mean is known and $n-1$ values are given, the n-th value can be computed.
- The dimension has changed, e.g. cm becomes cm^2.
- Comparison requires the same units. With different n, the precision of the estimates may differ.

Sample Variance – Worked Example

	Four A	Four B
Data	34, 37, 40, 41	36, 37, 39, 40
Mean $\bar{x}$	38	38

Calculation

$$s_A^2 = \frac{1}{4-1}\left[(34-38)^2 + (37-38)^2 + (40-38)^2 + (41-38)^2\right] = \frac{30}{3} = 10$$

$$s_B^2 = \frac{1}{4-1}\left[(36-38)^2 + (37-38)^2 + (39-38)^2 + (40-38)^2\right] = \frac{10}{3} \approx 3.33$$

Standard Deviation

Definition

The standard deviation is the square root of the variance:

$$s = +\sqrt{s^2} \qquad 0 \leq s \leq \infty$$

- By taking the square root, the measure of dispersion is again in the **dimension of the original values**
- If $s = 0$: no dispersion – all values equal the mean
- Same unit as the original values, therefore **more intuitive** than the sample variance
- Comparison requires the same units. With different n, the precision of the estimates may differ.

Standard Deviation – Worked Example

Calculation

$$s_A = +\sqrt{s_A^2} = +\sqrt{10} = 3.1623$$

$$s_B = +\sqrt{s_B^2} = +\sqrt{3.\overline{3}} = 1.8257$$

Interpretation

The values of Four A deviate on average by 3.16 units from the mean, while for Four B only by 1.83 units.

$\rightarrow$ The dispersion is now expressed in the **same unit** as the original data.

Coefficient of Variation

Definition

The coefficient of variation is a **relative** measure of dispersion:

$$v = \frac{s}{\bar{x}} \qquad \text{for data with } x_i > 0 \text{ for all } i \text{ or } x_i < 0 \text{ for all } i$$

- **Relative** means: independent of the scale
- The values must be either all positive or all negative
  - Mathematically: $\bar{x} \neq 0$, since otherwise division by zero
  - Substantively: $\bar{x}$ must reflect a meaningful **level** of the data
  - With mixed signs, $\bar{x} \approx 0$ is possible – then v becomes extremely large, even though the dispersion is not unusual
  - Example: $-10, -5, +5, +11 \Rightarrow \bar{x} = 0.25$, v loses its interpretive value
- However: with very different sample sizes, the precision of the estimates differs, so comparisons should be interpreted with caution

Coefficient of Variation – Worked Example

Calculation

$$v_A = \frac{s_A}{\bar{x}_A} = \frac{3.1623}{38} = 0.0832$$

$$v_B = \frac{s_B}{\bar{x}_B} = \frac{1.8257}{38} = 0.0480$$

Interpretation

Four A deviates relatively by about 8.3 % from the mean, Four B only by about 4.8 %.
$\rightarrow$ As a relative measure, v allows comparison of dispersions **regardless of the scale** of the data.

Normalized Coefficient of Variation

Definition

The normalized coefficient of variation is a **relative** measure of dispersion:

$$v^* = \frac{s}{\bar{x}} \cdot \frac{1}{\sqrt{n}} = \frac{v}{\sqrt{n}} \qquad \text{for data with } x_i > 0 \text{ for all } i \text{ or } x_i < 0 \text{ for all } i$$

- Same prerequisites as for the coefficient of variation v
- **Advantage over v:** By dividing by $\sqrt{n}$, v^* is **explicitly scaled by the sample size n**
- v^* thus combines both advantages:
  - Division by $\bar{x} \rightarrow$ independent of the **scale** (e.g. cm vs. km)
  - Division by $\sqrt{n} \rightarrow$ explicitly scaled by the **sample size**

Normalized Coefficient of Variation – Worked Example

Calculation

$$v_A^* = \frac{v_A}{\sqrt{n}} = \frac{0.0832}{\sqrt{4}} = \frac{0.0832}{2} = 0.0416$$

$$v_B^* = \frac{v_B}{\sqrt{n}} = \frac{0.0480}{\sqrt{4}} = \frac{0.0480}{2} = 0.0240$$

Interpretation

Both values are close to 0 $\rightarrow$ low relative dispersion.
$\rightarrow$ Because v^* is explicitly scaled by $\sqrt{n}$, these values are more comparable across different sample sizes than v alone.

Overview of Measures of Dispersion

	Four A	Four B
Mean	38	38
Median	37	37
Sample variance	10	3.3333
Standard deviation	3.1623	1.8257
Coefficient of variation	0.0832	0.0480
Normalized coefficient of variation	0.0416	0.0240

Measures of Dispersion with R

```r
# Variance
var(data)

# Standard deviation
sd(data)

# Coefficient of variation
v ← sd(data) / mean(data)

# Normalized coefficient of variation
vstar ← v / sqrt(length(data))
```

Exercise: Which Measure of Dispersion is Suitable for Which Comparison?

1. The shoe sizes of the rowing four and the rowing eight.
2. The shoe sizes of the women's four and the men's four.
3. The shoe sizes of the women's four and the women's four with coxswain.
4. The shoe sizes of the women's four and the men's eight.

Exercise: Measures of Dispersion – Explanation

Measure of dispersion	**Scale**	**Sample size n**
Sample variance s^2	dependent	precision depends on n
Standard deviation s	dependent	precision depends on n
Coefficient of variation v	independent	precision depends on n
Norm. coeff. of variation v^*	independent	explicitly scaled by n

Exercise: Measures of Dispersion – Solution 1

1. **Four vs. Eight** (same unit, same scale, **different** n)
   - s^2, s, v: less suitable, since different n (4 vs. 8) affects precision
   - v^*: **yes** – explicitly scaled by n
2. **Women's four vs. men's four** (same unit, **different scale**, same n)
   - s^2, s: less suitable, since men's shoe sizes are systematically larger
   - v: **yes** – not affected by scale, n is equal
   - v^*: **yes**

Exercise: Measures of Dispersion – Solution 2

3. Women's four vs. women's four with coxswain (same unit, same scale, **different** n)

- s^2, s, v: less suitable, since different n (4 vs. 5) affects precision
- v^*: **yes** – explicitly scaled by n

4. Women's four vs. men's eight (**different scale** and **different** n)

- s^2, s: less suitable, scale differs
- v: less suitable, different n affects precision
- v^*: **yes** – not affected by scale and explicitly scaled by n

Exercise: Which Measure of Dispersion is Suitable for Which Comparison?

1. The shoe sizes of the rowing four and the rowing eight.
2. The shoe sizes of the women's four and the men's four.
3. The shoe sizes of the women's four and the women's four with coxswain.
4. The shoe sizes of the women's four and the men's eight.

	1.	2.	3.	4.
Sample variance	no	no	no	no
Standard deviation	no	no	no	no
Coefficient of variation	no	yes	no	no
Normalized coefficient of variation	yes	yes	yes	yes

Exercise

Calculate the measures of dispersion for the shoe sizes of the handball team!

Exercise – Calculate Measures of Dispersion

Shoe sizes of the handball team ($n = 13$):

$$40, 39, 38, 41, 46, 39, 39, 38, 37, 40, 39, 40, 42$$

Calculate:

Measure	**Formula**	**Value**
Mean $\bar{x}$	$\frac{1}{n}\sum_{i=1}^{n} x_i$	?
Sample variance s^2	$\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$	?
Standard deviation s	$+\sqrt{s^2}$	?
Coefficient of variation v	$\frac{s}{\bar{x}}$	?
Norm. coeff. of variation v^*	$\frac{v}{\sqrt{n}}$	?

Exercise – Results

Shoe sizes: 40, 39, 38, 41, 46, 39, 39, 38, 37, 40, 39, 40, 42 $(n = 13)$

Measure of dispersion	Formula	Value
Mean $\bar{x}$	$\frac{1}{n}\sum x_i$	39.8462
Sample variance s^2	$\frac{1}{n-1}\sum(x_i - \bar{x})^2$	5.1410
Standard deviation s	$\sqrt{s^2}$	2.2674
Coefficient of variation v	$\frac{s}{\bar{x}}$	0.0569
Norm. coeff. of variation v^*	$\frac{v}{\sqrt{n}}$	0.0158

Overview

1 Fundamentals

2 Descriptive Statistics

3 Summary

Summary 1: Key Concepts

Fundamentals

- Statistical unit (ω): object being observed, defined by subject, time, location
- Variable (X): characteristic of interest; Observation (x_i): actually observed value
- Levels of measurement: nominal (categories) $\rightarrow$ ordinal (ranking) $\rightarrow$ metric (differences meaningful)
- Metric data is required for most statistical analyses in this course

Frequencies

Absolute frequency: $h_j = \#\{i \mid x_i = x_j\}$

Relative frequency: $f_j = \dfrac{h_j}{n}$

Cumulative relative frequency: $F(x) = \sum_{x_j \leq x} f_j$

Summary 2: Measures of Central Tendency

Measure	Formula	Properties
Arithmetic mean	$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$	Easy to compute; sensitive to outliers
Geometric mean	$\bar{x}_{\text{geo}} = \left(\prod_{i=1}^{n} x_i\right)^{1/n}$	Recommended for growth rates
Median	$x_{\lceil 0.5 \cdot n \rceil}$	Robust to outliers; also for ordinal data
Mode	Most frequent value	Also for nominal data; often not unique

Quantiles

For the p-quantile: position $= \lceil n \cdot p \rceil$ in the sorted data.
Quartiles: $Q_1 = x_{\lceil 0.25n \rceil}$, $Q_2 = x_{\lceil 0.5n \rceil}$ (Median), $Q_3 = x_{\lceil 0.75n \rceil}$

Summary 3: Measures of Dispersion

Measure	Formula	Key property
Variance s^2	$\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$	Dimension changes (e.g. cm^2)
Std. deviation s	$+\sqrt{s^2}$	Same unit as original data
Coeff. of variation v	$\frac{s}{\bar{x}}$	Scale-independent
Norm. coeff. v^*	$\frac{v}{\sqrt{n}}$	Scale-independent & scaled by n

Which measure for which comparison?

- Same scale, same n: s^2 or s sufficient
- Different scale, same n: use v
- Different n (with or without different scale): use v^*

Summary 4: Key R Commands (1/2)

Frequencies

```
table(data)                          # Absolute frequencies
prop.table(table(data))              # Relative frequencies
cumsum(prop.table(table(data)))      # Cumulative frequencies
```

Measures of Central Tendency

```
mean(data)                           # Arithmetic mean
mean(data, trim = 0.10)              # Trimmed mean (10%)
median(data)                         # Median
quantile(data, 0.25, type = 1)       # Quantiles (e.g. Q1)
```

Summary 4: Key R Commands (2/2)

Measures of Dispersion

```
var(data)                                # Sample variance
sd(data)                                 # Standard deviation
sd(data) / mean(data)                    # Coefficient of variation
sd(data)/mean(data)/sqrt(length(data))   # Norm. coeff. of var.
```

Outlook: Part 2 – Graphical Representations

What we have so far

The shoe sizes are now described by **numbers**: frequencies, measures of central tendency and measures of dispersion.

Next part

The same data are described by **pictures**: the empirical cumulative distribution function, the histogram and the boxplot – the quantiles of this part reappear as the building blocks of the boxplot.

Key takeaway

Numbers and pictures complement each other: a good description of data always uses both.